

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

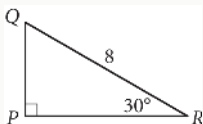
Right triangles and trigonometry

01

10 questions · ~15 min

Q1

GRID-IN GEOMETRY AND TRIGONOMETRY

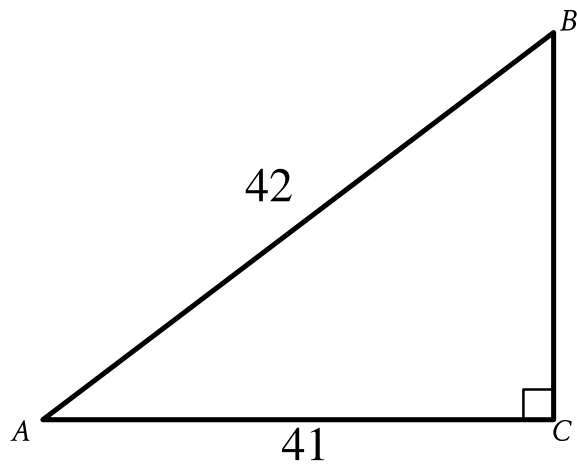


In the right triangle shown above, what is the length of $\overline{PQ}$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- Angle upper A upper C upper B is a right angle.
- The length of side upper A upper B is 42.
- The length of side upper A upper C is 41.
- A note indicates the figure is not drawn to scale.

What is the value of $\cos A$ in the triangle shown?

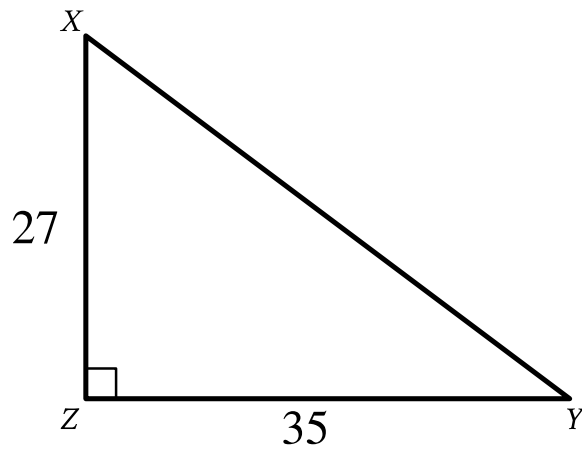
- A. $\frac{42}{41}$
- B. $\frac{41}{42}$
- C. $\frac{1}{42}$
- D. $\frac{1}{41}$

In right triangle RST , the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin R$ is $\frac{\sqrt{15}}{4}$. What is the value of $\cos S$?

- A. $\frac{\sqrt{15}}{15}$
- B. $\frac{\sqrt{15}}{4}$
- C. $\frac{4\sqrt{15}}{15}$
- D. $\sqrt{15}$

In $\triangle ABC$, $\angle B$ is a right angle and the length of $\overline{BC}$ is 136 millimeters. If $\cos A = \frac{3}{5}$, what is the length, in millimeters, of $\overline{AB}$?

- A. 34
- B. 102
- C. 136
- D. 170



Note: Figure not drawn to scale.

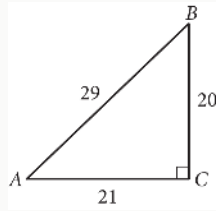
- Angle upper Z is a right angle.
- The length of side upper X upper Z is 27.
- The length of side upper Y upper Z is 35.
- A note indicates the figure is not drawn to scale.

Triangle XYZ shown is a right triangle. Which of the following has the same value as $\sin X$?

- A. $\tan X$
- B. $\tan Y$
- C. $\cos X$
- D. $\cos Y$

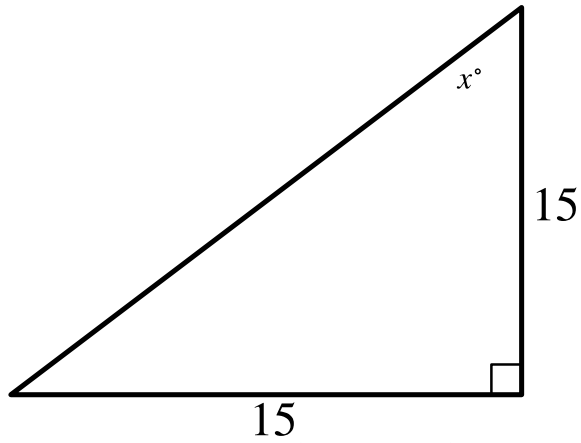
One leg of a right triangle has a length of 43.2 millimeters. The hypotenuse of the triangle has a length of 196.8 millimeters. What is the length of the other leg of the triangle, in millimeters?

- A. 43.2
- B. 120
- C. 192
- D. 201.5



In the figure above, what is the value of $\tan(A)$?

- A. $\frac{20}{29}$
- B. $\frac{21}{29}$
- C. $\frac{20}{21}$
- D. $\frac{21}{20}$



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x° .
- The lengths of the 2 sides adjacent to the right angle are as follows:
 - 15
 - 15
- Note: Figure not drawn to scale.

In the triangle shown, what is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8



Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A right triangle has legs with lengths of 28 centimeters and 20 centimeters. What is the length of this triangle's hypotenuse, in centimeters?

- A. $8\sqrt{6}$
- B. $4\sqrt{74}$
- C. 48
- D. 1,184

The length of a rectangle's diagonal is $5\sqrt{17}$, and the length of the rectangle's shorter side is 5. What is the length of the rectangle's longer side?

- A. $\sqrt{17}$
- B. 20
- C. $15\sqrt{2}$
- D. 400